Problem 4 | Problems T3_Diffusion.pdf #4 | Fick's 1st Law · Ceramic Barrier Transport

Ionic Transport of Nickel Through an MgO Ceramic Plate

Original statement · Problems T3_Diffusion.pdf

4.- A plate of magnesium oxide of $0.1\ \text{cm}$ *thickness separates two metallic blocks, one of* Ni *and one of* Ta*. At* 1400°C Ni *ions diffuse through the* MgO *plate. The diffusion coefficient of* Ni *in* MgO *is* $9 \times 10^{-12}\ \text{cm}^2/\text{s}$ *and the lattice parameter of (fcc)* Ni *at* 1400°C *is* $3.6 \times 10^{-8}\ \text{cm}$*. Determine the time necessary for sufficient ions* Ni^{2+} *ions to pass through the ceramic material so that the thickness of the block of* Ni *is reduced by one micron.* Official Solution: t = 309 h

PHASE 1 · HYPOTHESES & PARAMETERS

- **Quasi-Steady State:** Because the plate thickness ($\Delta x = 0.1\ \text{cm} = 1000\ \mu\text{m}$) is much larger than the layer consumed ($\Delta h = 1\ \mu\text{m}$), flux through MgO is in steady state [Slide 16].
- **Interfacial Boundary Conditions:** At the Ni/MgO interface, the concentration equals the atomic density of nickel metal: $C_1 = \rho_{\text{at, Ni}} = 4/a^3$ (FCC, as the statement gives). At the MgO/Ta interface, the nickel that arrives is taken as removed at once: $C_2 \approx 0$ (assumption of the model).
- **Parameters:** $\Delta x = 0.1\ \text{cm}$, $D = 9 \times 10^{-12}\ \text{cm}^2/\text{s}$, $a = 3.6 \times 10^{-8}\ \text{cm}$, $\Delta h = 1\ \mu\text{m} = 10^{-4}\ \text{cm}$.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

1. Atoms of Ni removed per unit area: $N/A = \rho_{\text{at, Ni}} \cdot \Delta h = \frac{4 \cdot \Delta h}{a^3}$.
2. Steady-state flux via Fick's First Law [Session 5 Slide 16]:

$$J = D\frac{\Delta C}{\Delta x} = D\frac{\rho_{\text{at, Ni}}}{\Delta x}$$

3. Conservation of mass equates flux to removal rate:

$$J = \frac{N/A}{t} \implies \frac{\rho_{\text{at, Ni}} \cdot \Delta h}{t} = D\frac{\rho_{\text{at, Ni}}}{\Delta x} \implies t = \frac{\Delta x \cdot \Delta h}{D}$$

The atomic density $\rho_{\text{at, Ni}}$ elegantly cancels out.

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

1. Direct Evaluation with General Formula:

$$\Delta x = 0.1\ \text{cm}, \quad \Delta h = 10^{-4}\ \text{cm}, \quad D = 9 \times 10^{-12}\ \text{cm}^2/\text{s}$$

$$t = \frac{(0.1\ \text{cm}) \times (10^{-4}\ \text{cm})}{9 \times 10^{-12}\ \text{cm}^2/\text{s}} = \frac{10^{-5}\ \text{cm}^2}{9 \times 10^{-12}\ \text{cm}^2/\text{s}} = \frac{1}{9} \times 10^{7}\ \text{s} \approx 1.1111 \times 10^{6}\ \text{s}$$

2. Conversion to Hours:

$$t = \frac{1.1111 \times 10^{6}\ \text{s}}{3600\ \text{s/h}} \approx \mathbf{308.64}\ \text{h} \approx \mathbf{309}\ \text{h}$$

3. Step-by-Step Atomic Confirmation:

$$\rho_{\text{at, Ni}} = \frac{4}{(3.6 \times 10^{-8}\ \text{cm})^3} = 8.5734 \times 10^{22}\ \text{atoms/cm}^3$$

$$N/A = (8.5734 \times 10^{22}) \times (10^{-4}) = 8.5734 \times 10^{18}\ \text{atoms/cm}^2$$

$$J = (9 \times 10^{-12}) \times \frac{8.5734 \times 10^{22}}{0.1} = 7.71605 \times 10^{12}\ \text{atoms/cm}^2 \cdot \text{s}$$

$$t = \frac{8.5734 \times 10^{18}}{7.71605 \times 10^{12}} = 1.1111 \times 10^{6}\ \text{s} \approx \mathbf{309}\ \text{h}$$

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

Dimensional Check: $[t] = \frac{\text{cm} \cdot \text{cm}}{\text{cm}^2/\text{s}} = \text{s}$. **Physical Insight:** Nearly 13 days (309 hours) at 1400°C are needed to remove 1 μm of nickel: diffusion is slower for charged species, in compact structures and in materials with high T_m [Slide 27]. At fixed concentrations the flux falls as $1/\Delta x$, so a thicker plate would take proportionally longer [Slide 16].